3/2 Valves and Cylinders

Press the button and the clamp closes.
Let go and it opens by itself.



National 5 Engineering Science

2019 N5 Engineering Science: scan the code and try these.

Q3(a) **Q3(b)**

Page 3. Answer in your jotter.

Scan for the paper

Marking scheme

2026 N5 Engineering Science: scan the code and try these.

Q14(c)

Page 29. Answer in your jotter.

Scan for the paper

Marking scheme

PART 1

The Cylinder and the 3/2 Valve

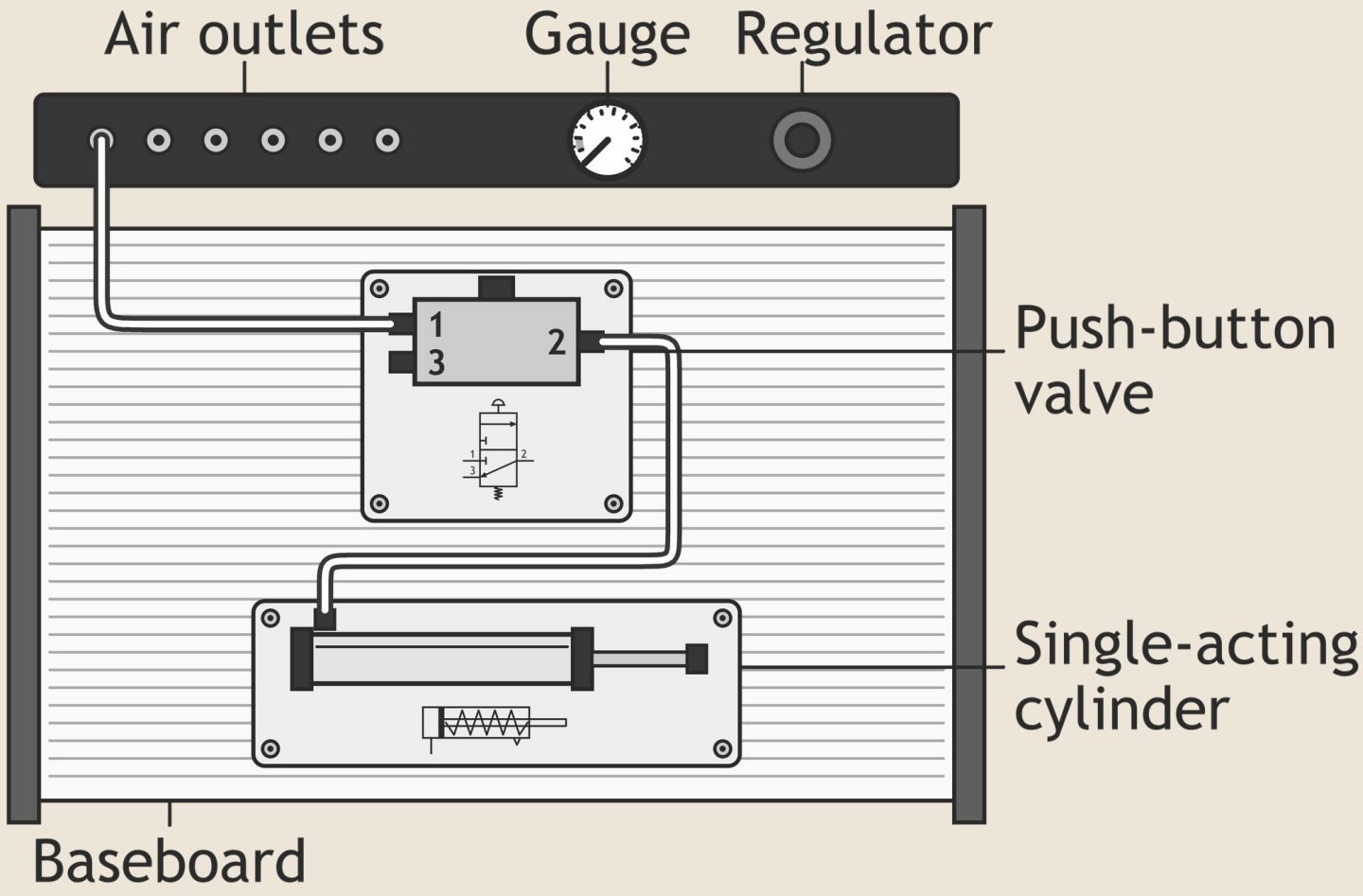
To describe how a 3/2 valve controls a single-acting cylinder, in exam words.

By the end of this part… (p.5)



1. I can **describe** how a single-acting cylinder outstrokes and instrokes.
2. I can **read** a 3/2 valve symbol: its ports, its two states, main air and exhaust.
3. I can **name** the ways a valve is actuated, from its symbol.
4. I can **describe** a 3/2 valve circuit using actuated, outstrokes and instrokes.

Let go of the button. What brings the piston back?

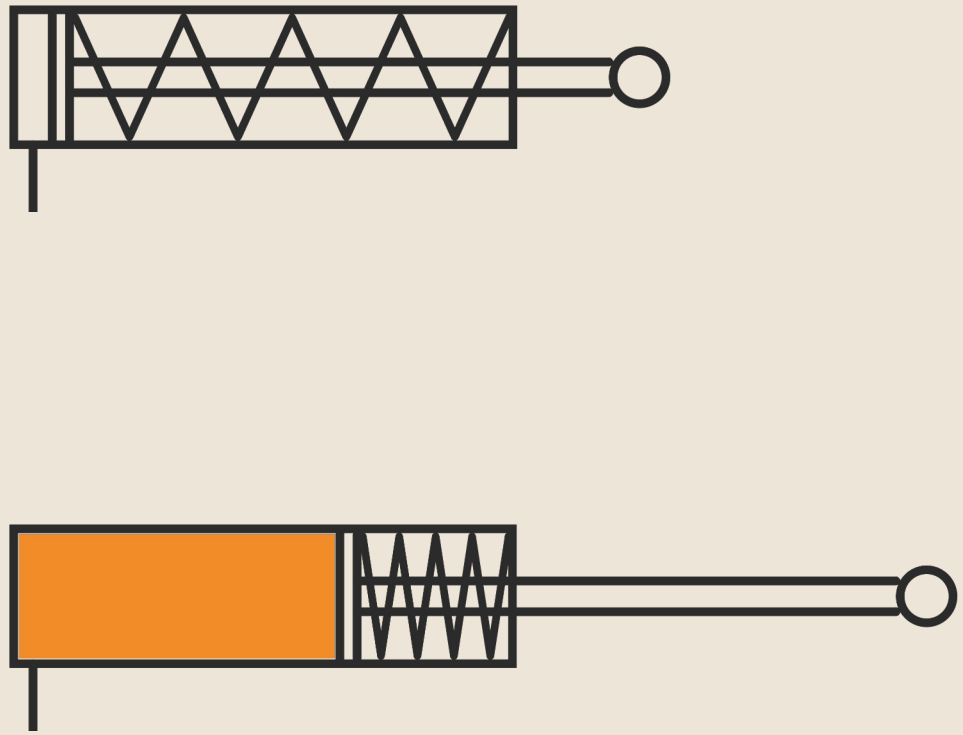


The Single-Acting Cylinder (p.5)

Main air pushes the piston, so the piston **outstrokes**.

When the air is exhausted, the **spring** pushes the piston back: it **instrokes**.

A single-acting cylinder is powered by air in **one** direction only.



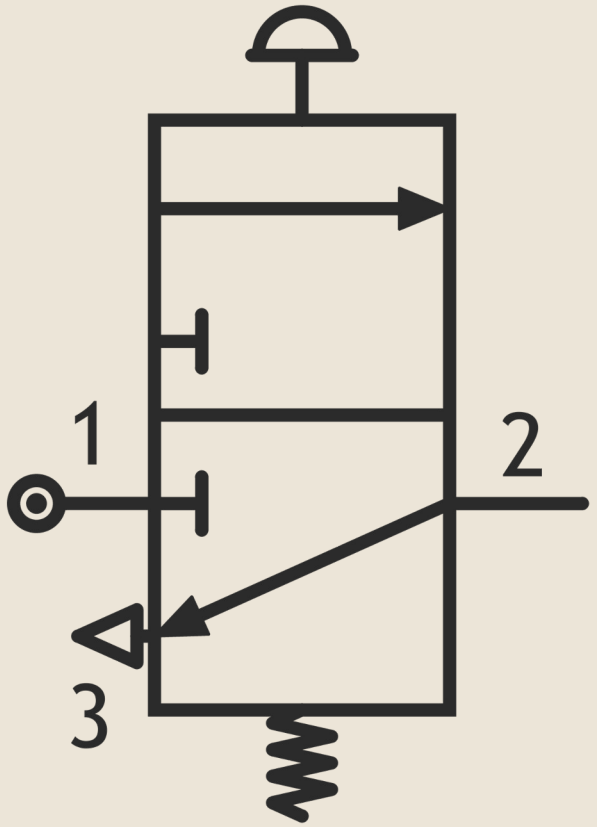
The 3/2 Valve (p.5)

3/2 means **3** ports and **2** states.

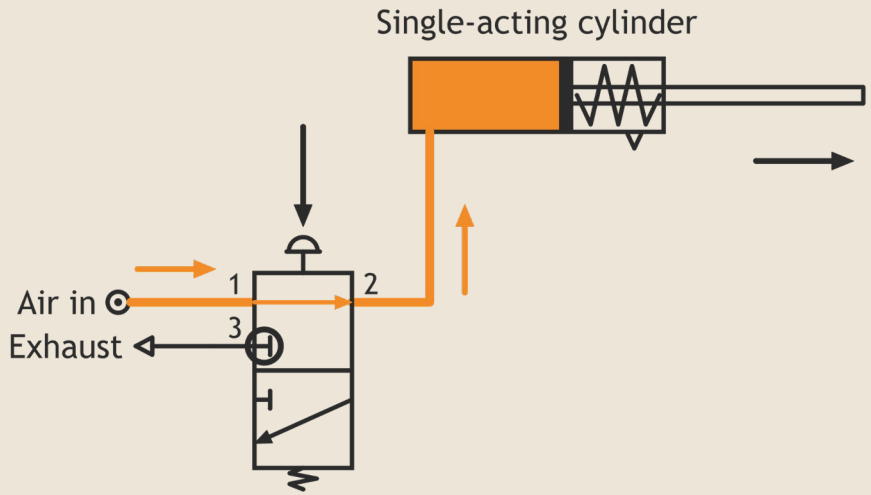
Port 1 is **main air**, port 2 goes to the **cylinder** and port 3 is **exhaust**.

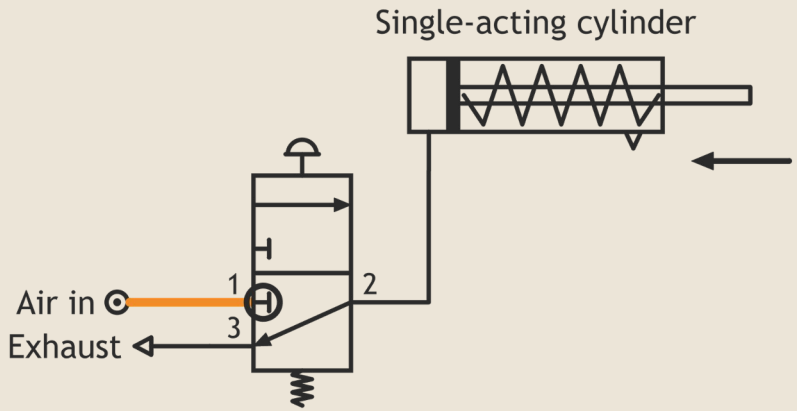
Each box shows one state. The box beside the spring is the **unactuated** state.

The spring under the valve returns it to the **unactuated** state when the actuator is released.



Actuated and Unactuated (p.5)

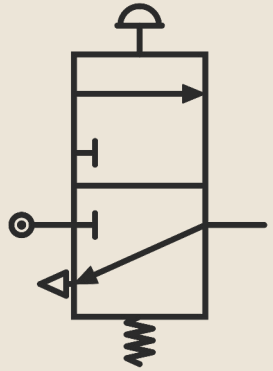




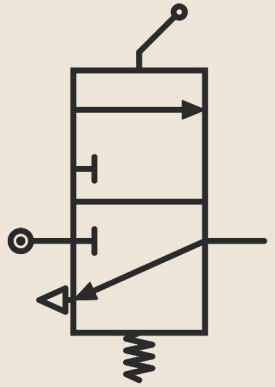
Actuated: air flows from port 1 to port **2**, so the piston outstrokes.

Unactuated: air flows from port 2 to port **3**, so the air exhausts and the piston instrokes.

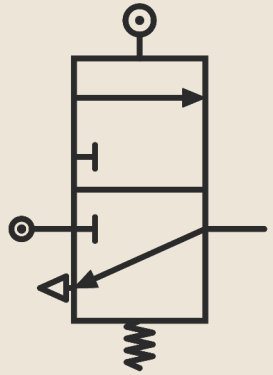
Ways to Actuate a Valve: People and Parts (p.6)



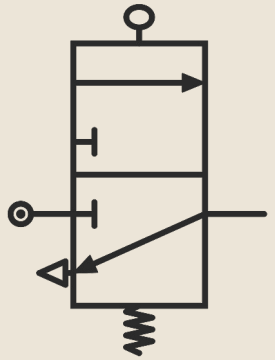
Push button
a person **presses** it



Lever
a person **moves** it

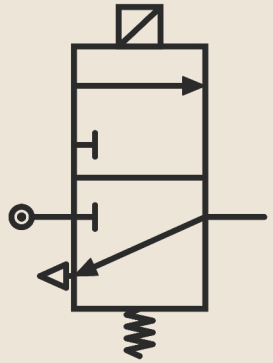


Roller
a moving part **rolls** over it

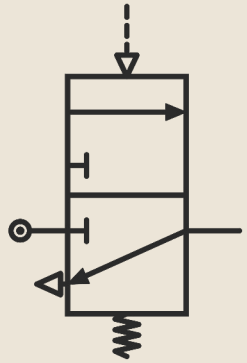


Plunger
a moving part **pushes** it

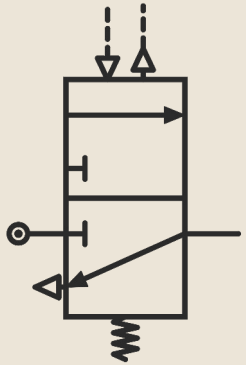
Ways to Actuate a Valve: Signals (p.6)



Solenoid
an **electrical** signal, e.g. from a microcontroller



Pilot air
an **air** signal from another valve

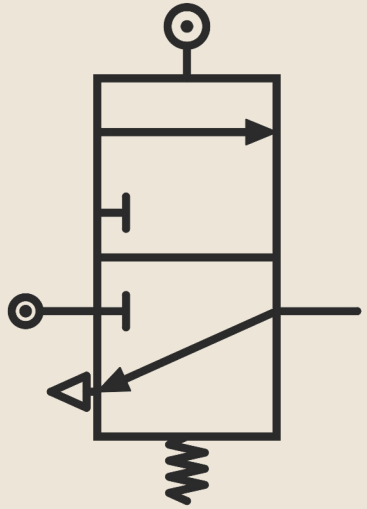


Diaphragm
a change in air **pressure**, e.g. an air bleed being covered

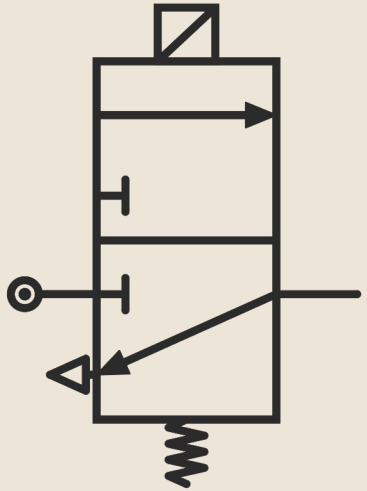
Show Me: Name the Actuator

Name each actuator, 1 to 4.

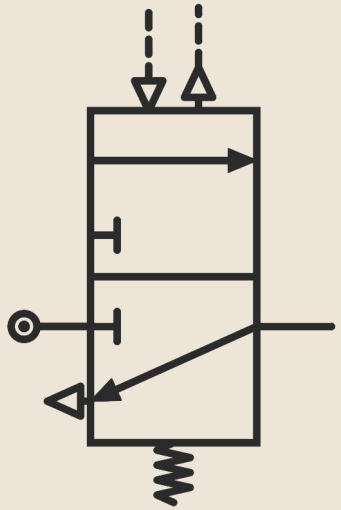
1



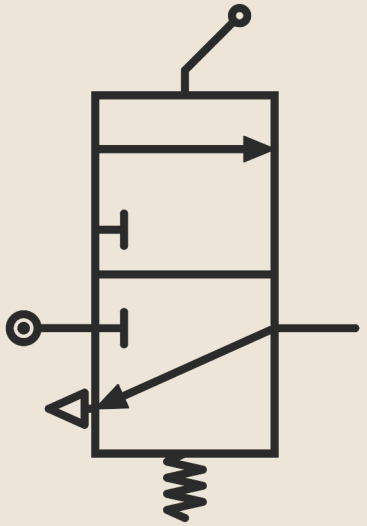
2



3



4



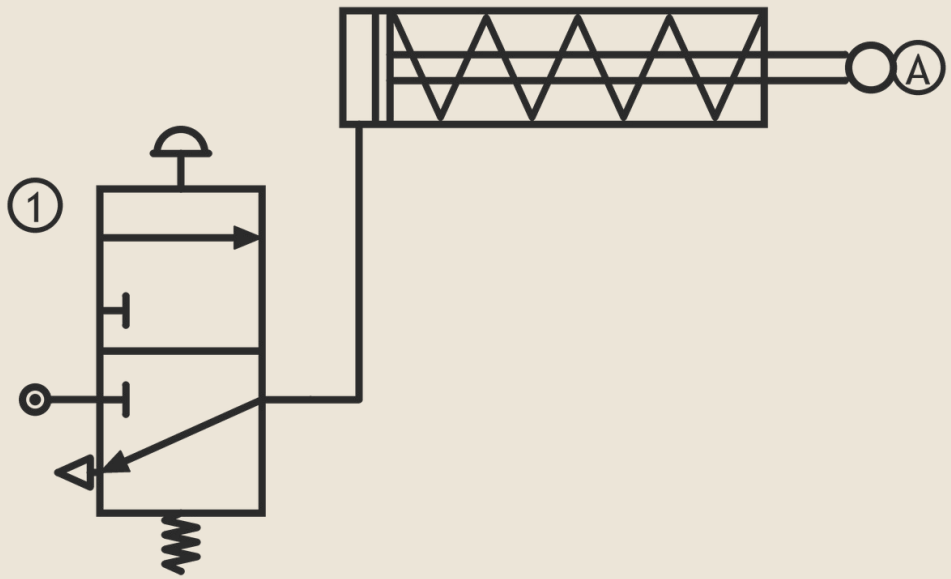
Words That Get the Marks (p.7)

When you describe a circuit, use

the valve is **actuated** (or unactuated)

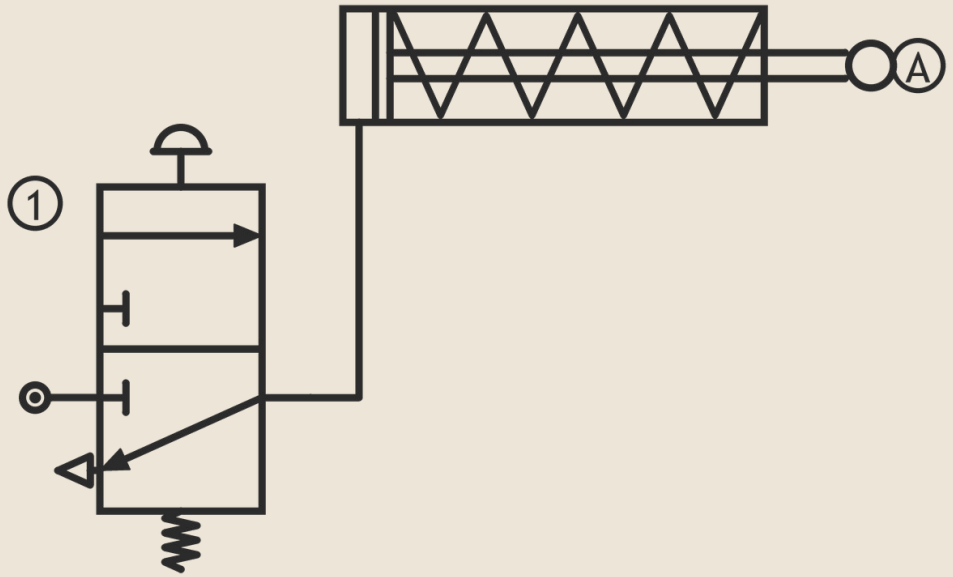
the piston **outstrokes** or **instrokes**

name every valve and piston by its number or letter



Model Answer (p.7)

When the push button is pressed, valve 1 is **actuated**. Main air flows to the cylinder, so the piston **outstrokes**. When the button is released, the spring returns valve 1 and the air **exhausts**. The spring in the cylinder makes the piston **instroke**.

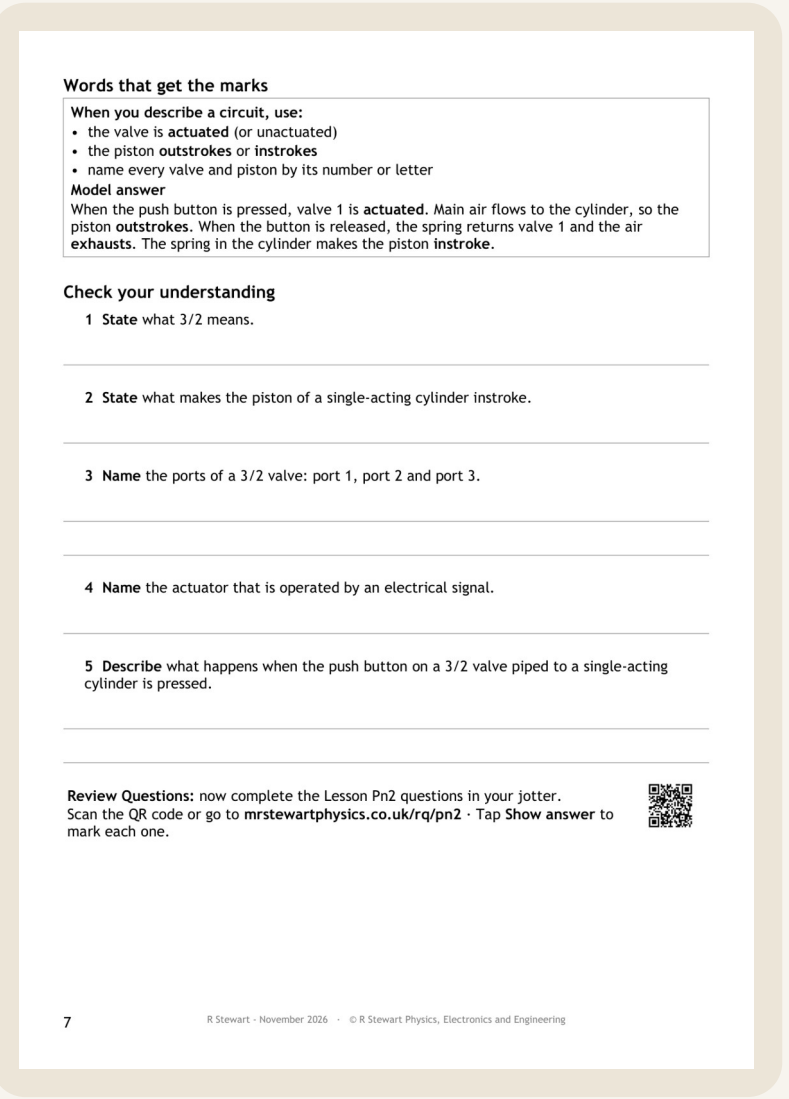


Check Your Understanding (pp 5-7)

In your booklet

Answer the **Quick checks** on pages **5** and **6**.

Answer Questions **1** to **5** on page **7**.



Words that get the marks

When you describe a circuit, use:
- the valve is **actuated** (or unactuated)
- the piston **outstrokes** or **instrokes**
- name every valve and piston by its number or letter

Model answer
When the push button is pressed, valve 1 is **actuated**. Main air flows to the cylinder, so the piston **outstrokes**. When the button is released, the spring returns valve 1 and the air **exhausts**. The spring in the cylinder makes the piston **instroke**.

Check your understanding

1 **State** what 3/2 means.

2 **State** what makes the piston of a single-acting cylinder instroke.

3 **Name** the ports of a 3/2 valve: port 1, port 2 and port 3.

4 **Name** the actuator that is operated by an electrical signal.

5 **Describe** what happens when the push button on a 3/2 valve piped to a single-acting cylinder is pressed.

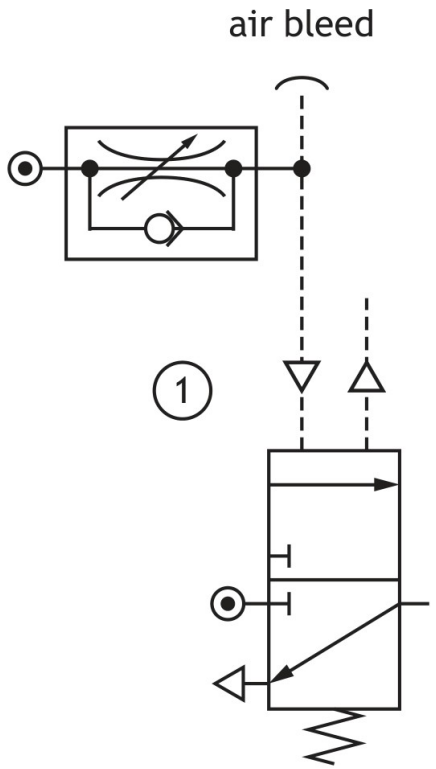
Review Questions: now complete the Lesson Pn2 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn2** · Tap **Show answer** to mark each one.

7

Answers: Check Your Understanding (pp 5-7)

Question	Answer
Quick check p.5	The **spring**.
Quick check p.6	A **roller** (or a plunger).
1	**3** ports and **2** states.
2	The **spring**.
3	1: **main air**. 2: to the **cylinder**. 3: **exhaust**.
4	A **solenoid**.
5	Valve **actuated** (1). Piston **outstrokes** (1).

Exam: 2026 Section 2 Question 14 (a)(ii)



(ii) State the name of the actuator that is operated by the signal from the air bleed. **1**

Question 14 (a)(ii)

1 mark

Page 28 of the paper.

Marking: 2026 Section 2 Question 14 (a)(ii)

What Candidates Get Wrong

Name the actuator

the **diaphragm** (2026 report, 43 %)

Pipe a 3/2 valve

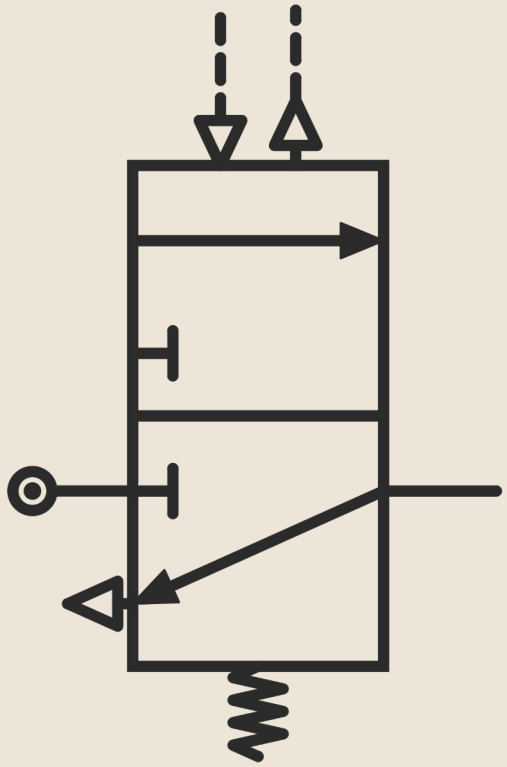
port to port (2024 Q3: 45 %)

Draw a solenoid

a **rectangle** and **inclined line** (2018 MI)

Describe a circuit

actuated, **outstrokes**, **instrokes** (2025)



Review Questions (p.7)

Lesson Pn2 (online)

Answer the Lesson **Pn2** questions in your **jotter**.
Scan the QR or go to **mrstewartphysics.co.uk/rq/pn2**.
Tap **Show answer** to **mark** each one.

Scan me

Look back — by the end of this part… (p.5)

1. I can **describe** how a single-acting cylinder outstrokes and instrokes.
2. I can **read** a 3/2 valve symbol: its ports, its two states, main air and exhaust.
3. I can **name** the ways a valve is actuated, from its symbol.
4. I can **describe** a 3/2 valve circuit using actuated, outstrokes and instrokes.

2021 N5 Engineering Science: scan the code and try these.

Q3(a)

Page 3. Sketch it in your jotter.

Scan for the paper

Marking scheme

PART 2

Build It: the Push-Button Clamp

To plan, simulate, build and evaluate a pneumatic circuit, as the assignment asks.

By the end of this part… (pp 8-9)



1. I can **write** a test plan with the expected result of each test.
2. I can **simulate** a circuit, then **build** it on the kit.
3. I can **evaluate** a circuit against its specification, using my tests.
4. I can **describe** a modification and **justify** it.

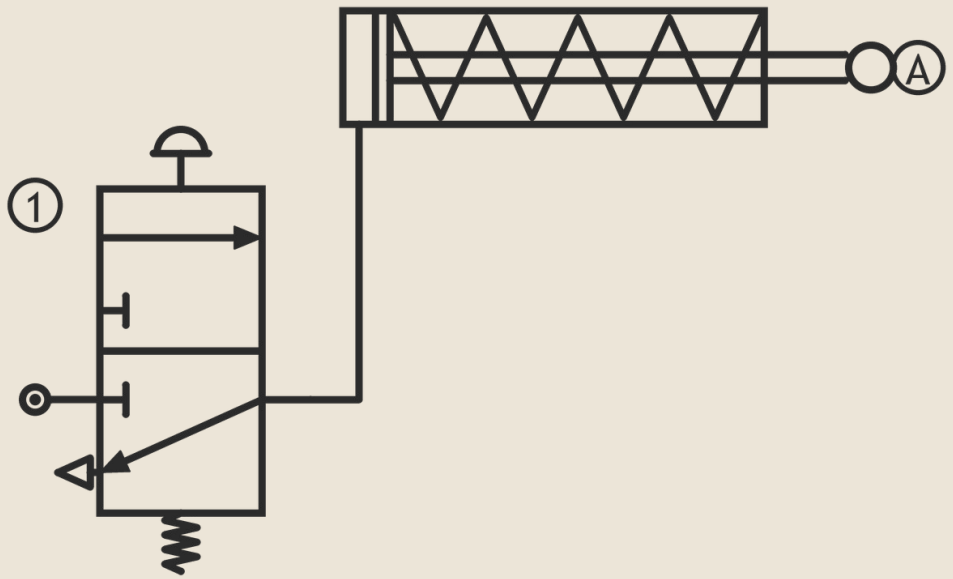
Build It: the Push-Button Clamp (p.8)

A clamp holds a part still while it is drilled. A push button outstrokes the piston.

A The piston **outstrokes** when the push button on valve 1 is pressed.

B The piston **instrokes** when the push button is released.

C The system must be **repeatable**.



Write a Test Plan (p.8)

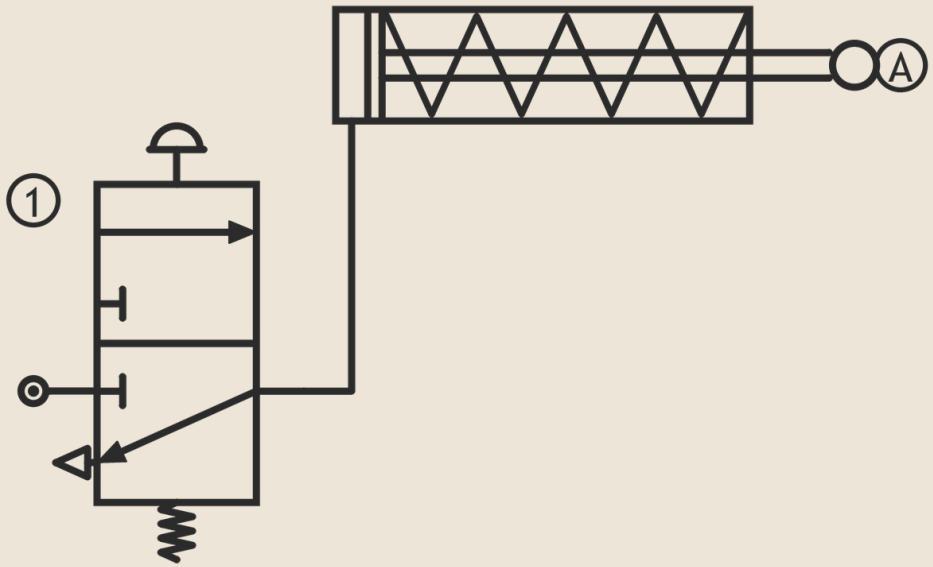


Planned test	Expected result	Test result
Press the push button on valve 1.		
Release the push button.		
Repeat tests 1 and 2 three times.		

Simulate It (p.8)

In your booklet

1. **Open** NoPressureSim. **Build** the circuit, main air on every valve.
2. **Predict**, then test. Were you right?
3. Air off: **build** it on the kit. Air on: **test**.

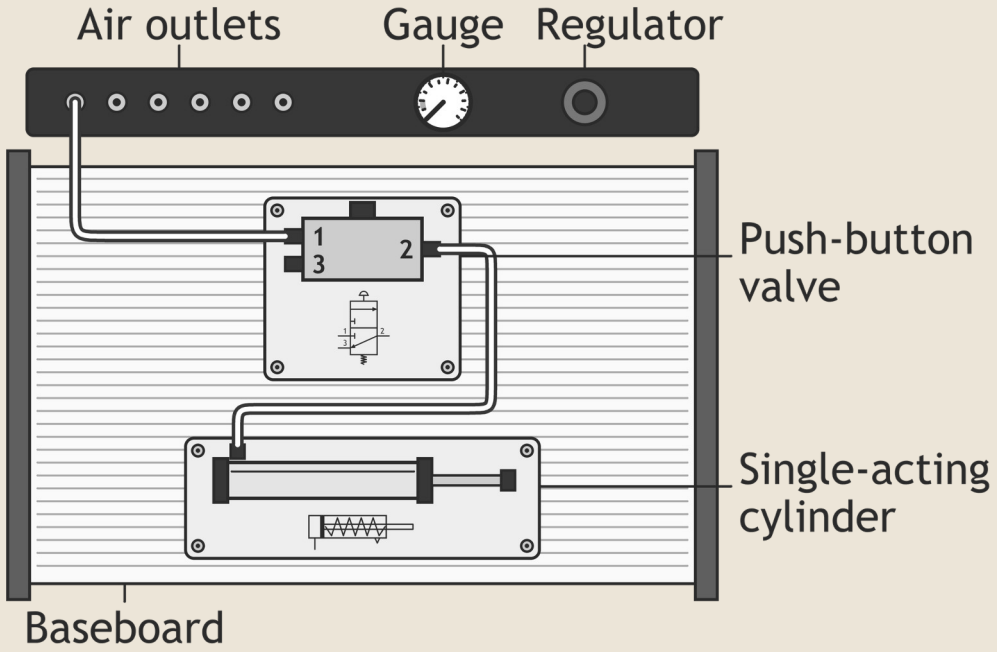


Scan me
NoPressureSim

What to Collect

Collect one set per pair.

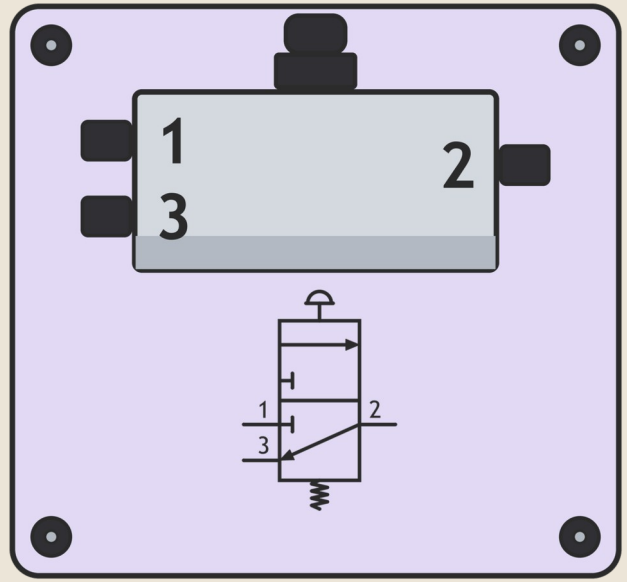
1. 3/2 push-button spring-return valve
2. Single-acting cylinder
3. Two lengths of blue tube
4. Goggles, one pair each



Build It on the Kit (p.8)

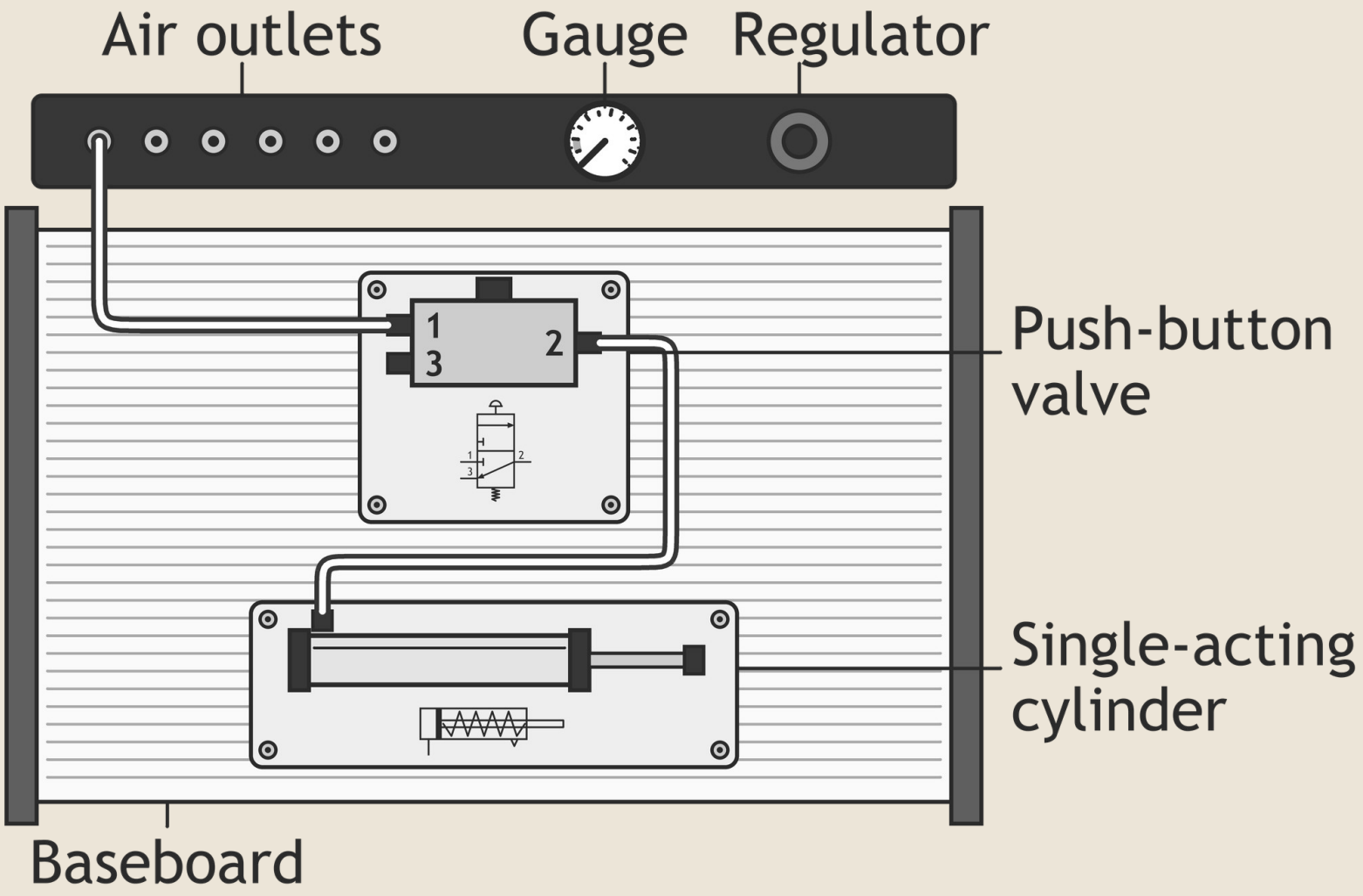
On the kit

Build the circuit, air **off**.
Pipe an air outlet to port **1**.
Pipe port **2** to the cylinder.
Air on. **Test** and **record** each result.



SAFETY 0 bar to build. 1 to 2 bar to test.

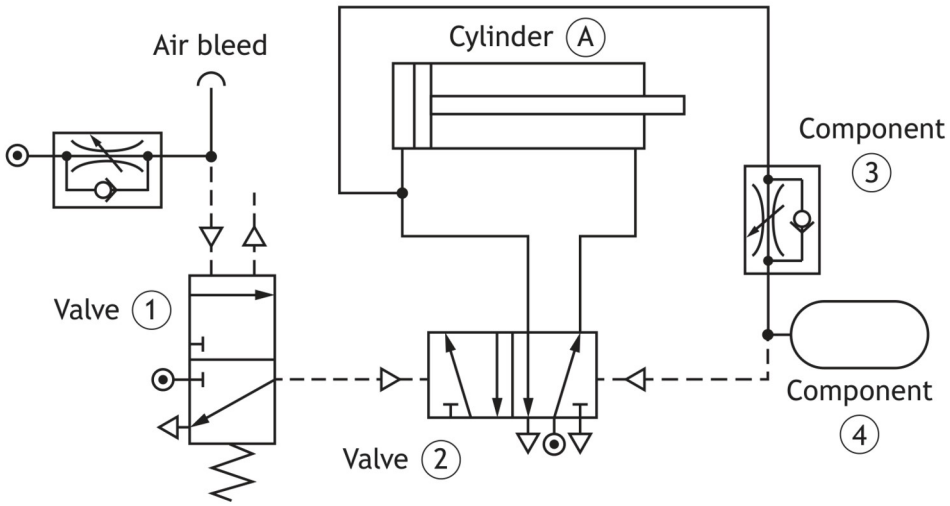
Build It: Bench Card 1



Exam: 2018 Section 2 Question 15 (d)

Portions of the pastry travel along a conveyor belt where a pneumatic piston presses them into pie casings.

The pneumatic circuit shown below operates the piston when the pastry is sensed in position.



(d) Explain why an air bleed was selected as an appropriate way of sensing the pastry. **2**

_____

_____

Marking: 2018 Section 2 Question 15 (d)

How It's Marked: Cause and Effect

1 mark: the cause

it is **clean**, or it does **not apply pressure**

1 mark: the effect

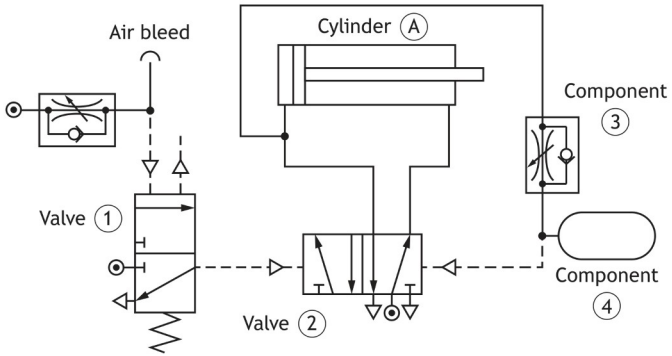
so it will not **contaminate** or **damage** the pastry

In context

the MI: answers must **refer to the pastry**

Portions of the pastry travel along a conveyor belt where a pneumatic piston presses them into pie casings.

The pneumatic circuit shown below operates the piston when the pastry is sensed in position.

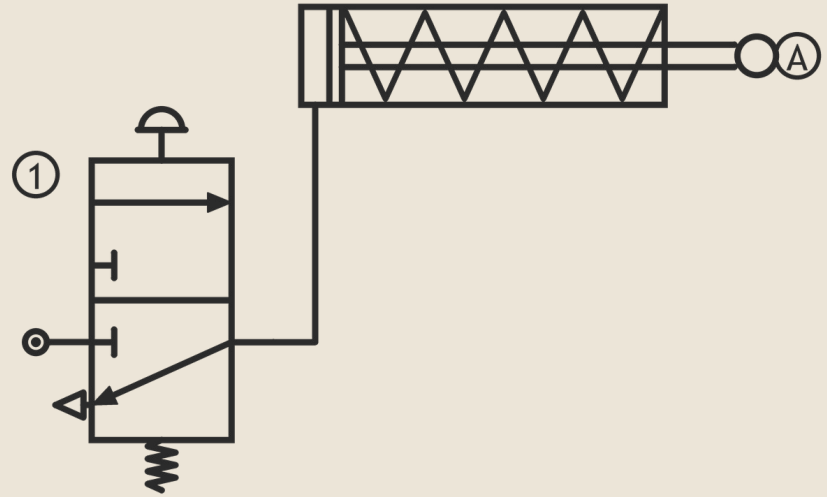


(d) Explain why an air bleed was selected as an appropriate way of sensing the pastry. **2**

Evaluate (p.9)

In your booklet

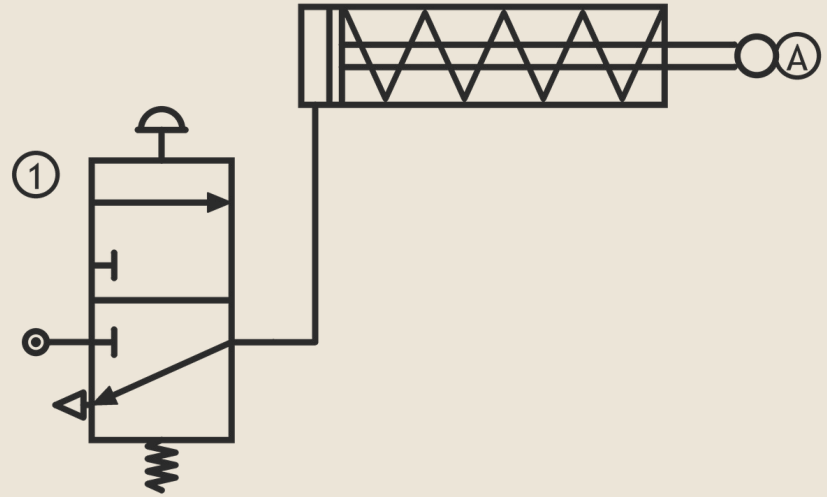
For each specification point, **state** if it was met.
Explain why, using your **test results**.
Say what you did **and** what the piston did.



Improve and Justify (p.9)

In your booklet

The operator's hand is **close to the clamp** when they press the button.
Describe a modification that would solve this.
Justify it: say **why** it solves the problem.



Answers: Evaluate (p.9)

Point	Met?	Explanation
A	Yes	**In test 1**, the button was **pressed**, valve 1 was **actuated**, **so** piston A **outstroked**.
B	Yes	**In test 2**, the button was **released**, the spring returned valve 1, **so** piston A **instroked**.
C	Yes	**In test 3**, piston A outstroked and instroked **every time**, over three repeats.

Answers: Improve and Justify (p.9)



	Answer
Modification	Use **two** push-button valves in **series**, one for each hand, away from the clamp (a **two-hand control**).
Justification	Both hands must press a button, **so** neither hand can be near the clamp when it closes.

What Candidates Get Wrong

Evaluate with cause and effect

what you did, **so** what it did (2026 report)

Show the air supply

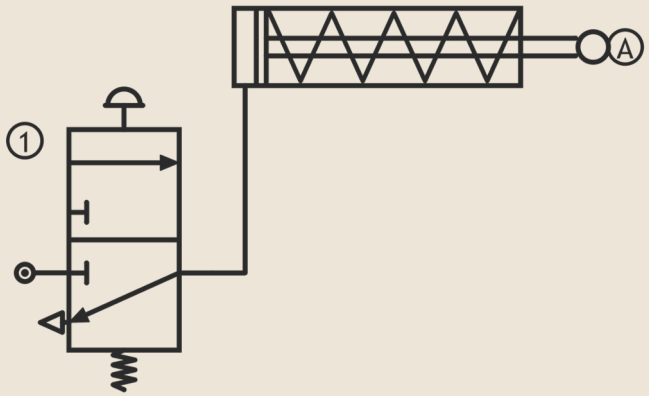
on **every** valve (2025 report)

Fix the stated issue

it must solve the **stated issue** (2025 report)

Explain in context

name the **product** (2018 Q15(d) MI)



Look back — by the end of this part… (pp 8-9)



1. I can **write** a test plan with the expected result of each test.
2. I can **simulate** a circuit, then **build** it on the kit.
3. I can **evaluate** a circuit against its specification, using my tests.
4. I can **describe** a modification and **justify** it.

Scan: page 6

Pneumatics

- symbols and operation of standard pneumatic components (restrictor, uni-directional restrictor, reservoir, 5/2 valve and actuators: diaphragm and solenoid)
- pneumatic time delay circuits
- calculating relationships between force, pressure and area in single-acting and double-acting cylinders
- controlling speed and force

Credits (1 of 3)

Title: Photo: Mixabest, CC BY-SA 3.0, via Wikimedia Commons

Starter: 2019 N5 Engineering Science, © Qualifications Scotland (SQA). 2019 question paper and marking instructions; 2026 N5 Engineering Science, © Qualifications Scotland (SQA). 2026 question paper and marking instructions; 2021 N5 Engineering Science, © Qualifications Scotland (SQA). 2021 question paper and marking instructions

Drawings: Pneumatic symbols and circuits drawn by R Stewart in the Qualifications Scotland paper style; CP2080 kit drawings by R Stewart.

Exam: 2026 Section 2 Question 14 (a)(ii): © Qualifications Scotland (SQA). 2026 National 5 Engineering Science Section 2, Q14

Marking: 2026 Section 2 Question 14 (a)(ii): © Qualifications Scotland (SQA). 2026 National 5 Engineering Science Section 2, Q14 marking instructions

Credits (2 of 3)

What Candidates Get Wrong: © Qualifications Scotland (SQA). National 5 Engineering Science course reports 2025 and 2026; 2024 S1 Q3; 2018 S1 Q7 marking instructions; © Qualifications Scotland (SQA). National 5 Engineering Science course reports 2025 and 2026 (assignment); 2018 S2 Q15(d) marking instructions

Screenshots and page figures: Booklet 6, Pneumatics (N5 Engineering Science, R Stewart, November 2026): page images.

Build It: the Push-Button Clamp: © Qualifications Scotland (SQA). 2024 National 5 Engineering Science Section 1, Q3

Exam: 2018 Section 2 Question 15 (d): © Qualifications Scotland (SQA). 2018 National 5 Engineering Science Section 2, Q15

Marking: 2018 Section 2 Question 15 (d): © Qualifications Scotland (SQA). 2018 National 5 Engineering Science Section 2, Q15 marking instructions

Credits (3 of 3)

Course specification: © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.6

R Stewart - November 2026